

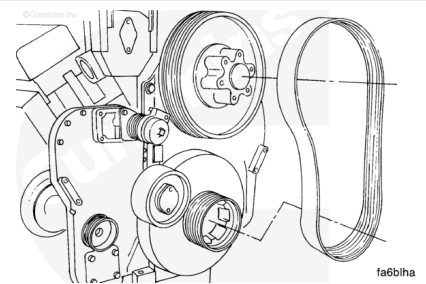
Trainings ▾

Remove

Shock Absorber System

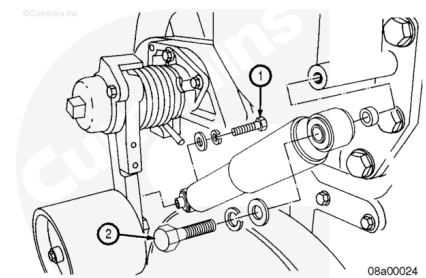
Loosen the fan belt tension. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/57/57-008-002.html>).

Remove the fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/57/57-008-002.html>).



Remove the backside idler pulley end of the shock absorber.

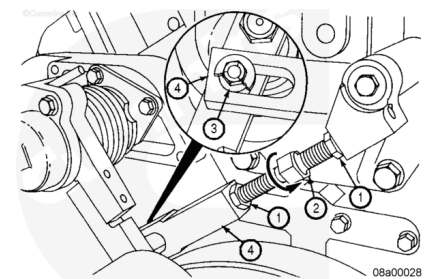
Loosen the upper capscrew (2). Remove the lower capscrew (1).



Turnbuckle Idler System

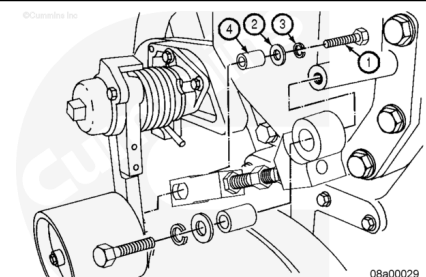
Note : One of the jam nuts on the solid control rod (turnbuckle) has left-handed threads.

Loosen the solid control rod (turnbuckle) jam nuts (1). Turn the adjusting screw (2) until the spacer (3) is **not** touching the end of the slot in the control rod (4).



Remove the capscrew (1), washers (2, 3), and spacers (4).

Remove the control rod assembly from the idler assembly.

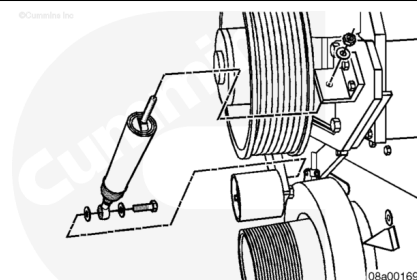


Industrial System

Remove the capscrews and washers from the lower clevis and idler arm.

Remove the nut and the capscrew from the threaded rod end of the tensioner at the upper anchor bracket.

Remove the industrial fan belt tensioner.

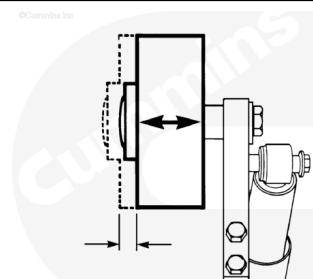


Inspect for Reuse

Power Generation

Check the clearance.

mm		in
0.08	MIN	0.003
0.25	MAX	0.010



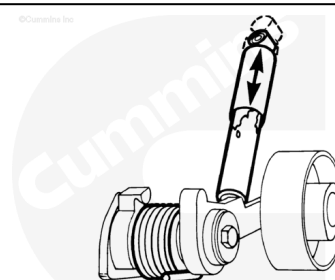
Rotate the pulley. Check for rough or damaged bearings.

The bearings **must** be replaced if any of the above conditions are found.

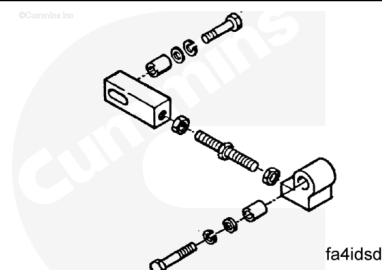
Check the shock absorber for cracks and fluid leakage.

Push the shock absorber up and down. Check the vibration absorption.

The shock absorber **must** be replaced if any of the above conditions are found.

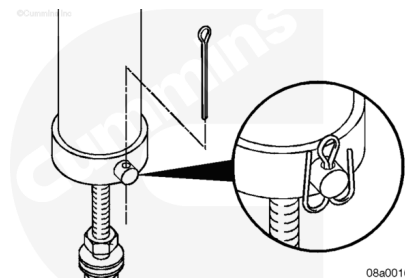


Note : Some engines have a turnbuckle instead of a shock absorber. Check the parts of the turnbuckle for damage.



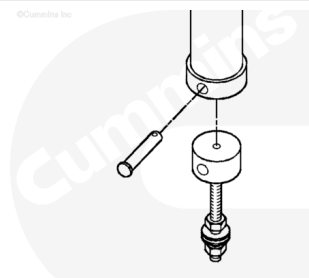
Industrial System

Remove the cotter pin from the clevis pin.



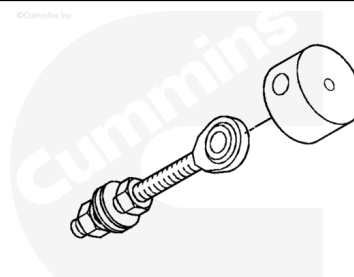
08a00167

Remove the clevis pin from the control rod and belt tensioner housing and separate.



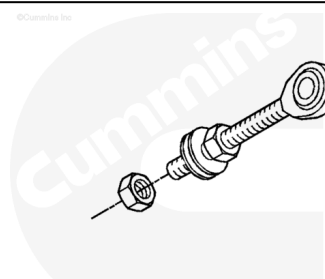
08a00166

Remove the control rod from the control rod retainer.



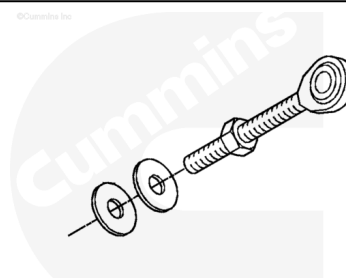
08a00165

Remove the first nut from the control rod threaded end.



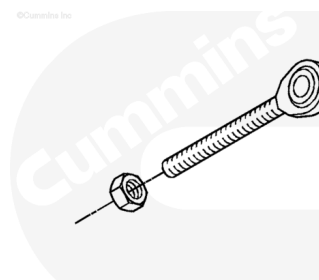
08a00164

Remove the two flat washers from the control rod threaded end.



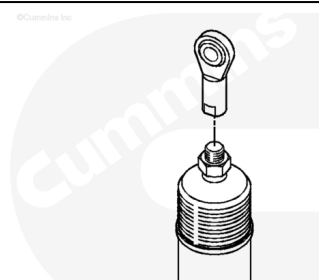
08a00163

Remove the remaining nut from the control rod threaded end.



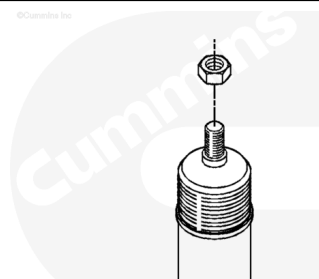
08a00162

Remove the control rod from the capscrew threaded end.



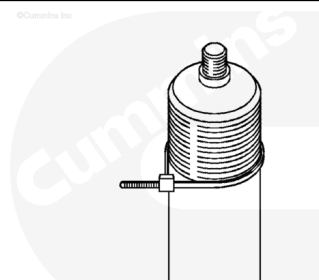
08a00161

Remove the nut holding the dust seal to the capscrew.



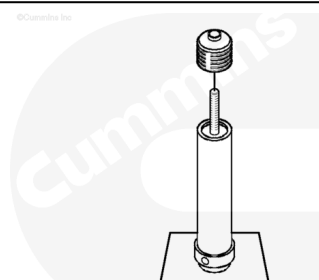
08a00160

Cut the wire tie to remove the dust seal from the belt tensioner housing.



08a00159

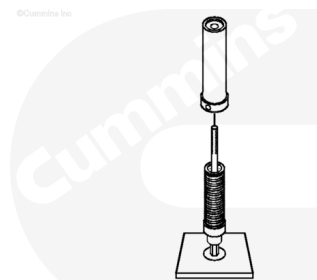
Remove the dust seal from the threaded end of the capscrew and belt tensioner housing.



08a00158

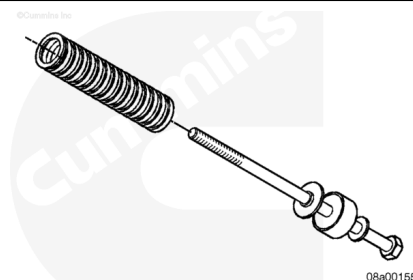
Install complete unit onto the build fixture with the head of the capscrew and remove the belt tensioner housing from the belt tensioner assembled parts.

Note : It is **not** necessary to use a build fixture to disassemble or assemble as shown in the picture. This is an aid to clarify the steps in the procedures.



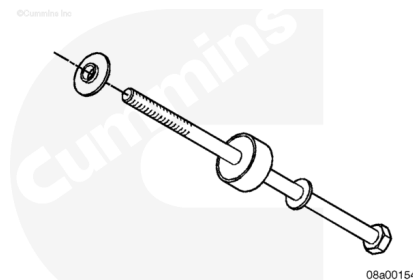
08a00157

Remove the compression spring from the spring retainer.



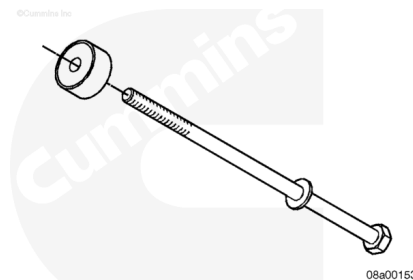
08a00155

Remove the spring retainer from the spring guide.



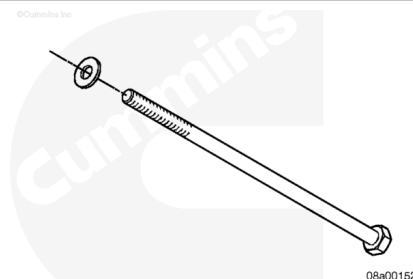
08a00154

Remove the spring guide from the flat washer.



08a00153

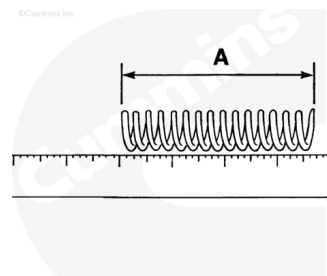
Remove the flat washer from the capscrew.



08a00152

Measure the free length of the spring

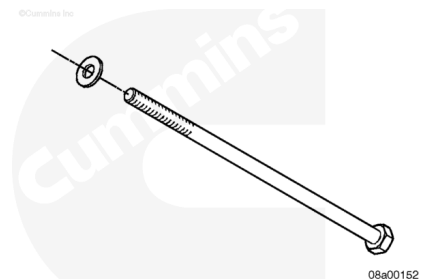
mm		in
147.3	MIN	5.80
155.4	MAX	6.12



kn8sp1d

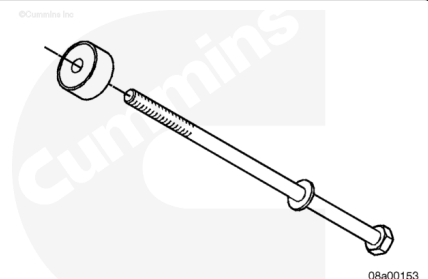
Note : Replace with new springs if not within specifications.

Install the flat washer onto the capscrew.



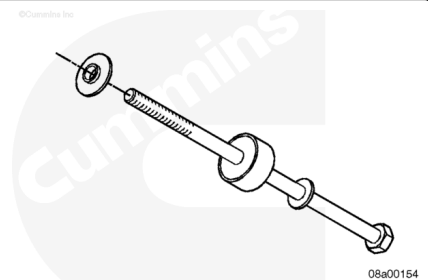
08a00152

Install the spring guide with the recessed/indented side next to the flat washer.



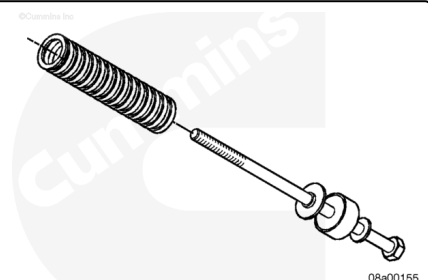
08a00153

Install the spring retainer with the large flat surface against the spring guide.



08a00154

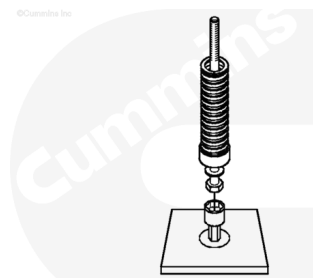
Install the compression spring against the spring retainer.



08a00155

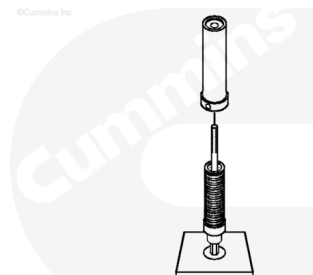
Set the assembled parts in the small build fixture with the head of the capscrew setting in the socket on the top of the build fixture.

Note : It is **not** necessary to use a build fixture to disassemble or assemble as shown in the picture. This is an aid to clarify the steps in the procedures.



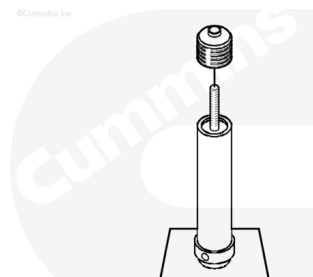
08a00156

Set the belt tensioner housing over the assembled parts and allow the bottom of the housing to set on the build fixture with the threads of the capscrew protruding out of the hole in the top of the belt tensioner housing.



08a00157

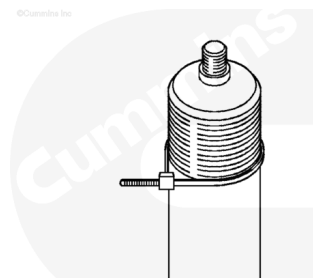
Slip the dust seal over the end of the capscrew and the edge of the belt tensioner housing.



08a00158

Use a wire tie and fasten around the dust seal where it overlaps the belt tensioner housing.

Pull the tie until tight and cut off the excess of the tie.

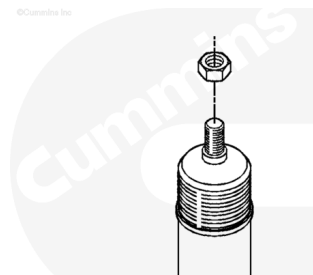


08a00159

Push the dust seal down over threads of the capscrew and install the nut.

Use a 3/4-inch open-end wrench to tighten the nut.

Note : Be sure just to tighten the nut far enough to get clearance for the following part to be installed. Do not completely tighten at this time.

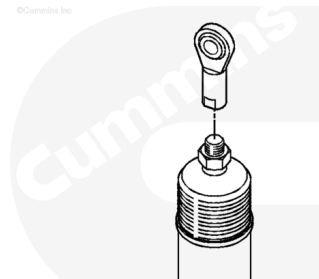


08a00160

Install the control rod end on the threaded end of the capscrew.

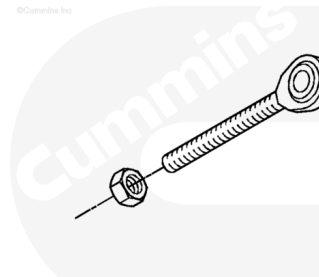
Back the nut against the control rod and tighten.

Torque Value: 100 n•m [74 ft-lb]



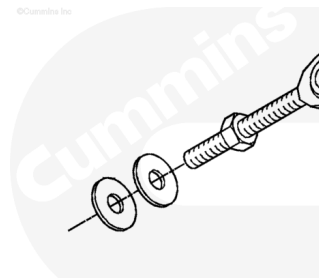
08a00161

Install the nut and install on the control rod end.



08a00162

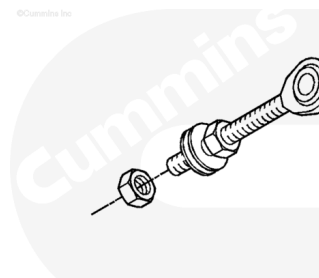
Install two flat washers onto the control rod end.



08a00163

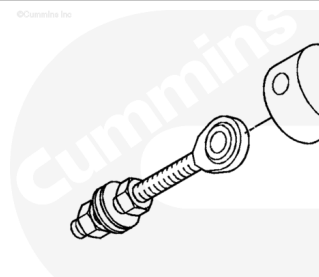
Install a second nut onto the control rod end.

Note : Do not tighten completely. Leave the washers loose.



08a00164

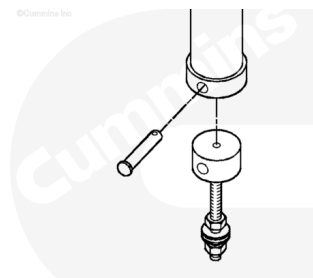
Install the assembled control rod end into the control rod retainer's slotted end.



08a00165

Install the assembled control rod parts into the end of the belt tensioner housing.

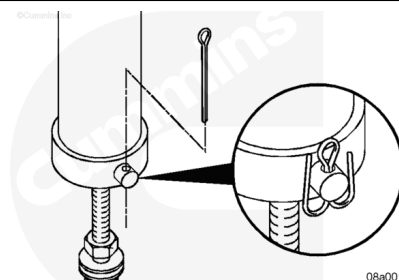
Align the holes and install the clevis pin to hold the parts together.



08a00166

Use a cotter pin to secure the clevis pin to the housings.

Note : Bend the cotter pin around the clevis pin to keep from falling out during operation.

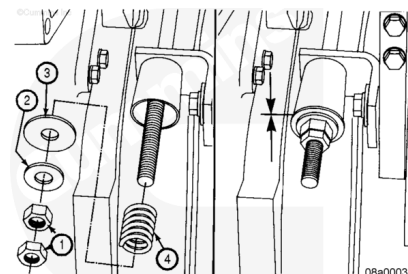


08a00167

Install

Power Generation

The backside idler system uses a control rod with a spring (4) held by retaining washers (2, 3) and jam nuts (1) or a shock absorber.



08a00030

Shock Absorber System

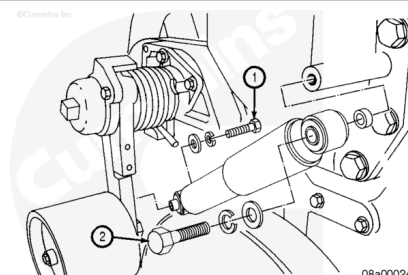
Install the fan hub assembly end of the shock absorber.

Tighten the upper capscrew (2).

Torque Value: 52 n•m [38 ft-lb]

Install and tighten the lower capscrew for the idler pulley end of the shock absorber (1).

Torque Value: 47 n•m [35 ft-lb]



08a00024

Turnbuckle Idler System

To install the control rod with spring, install the flat washer (3), lock washer (4), and capscrews (5) in the upper end of the control rod (2).

Install the control rod in the fan support.

Tighten the capscrew.

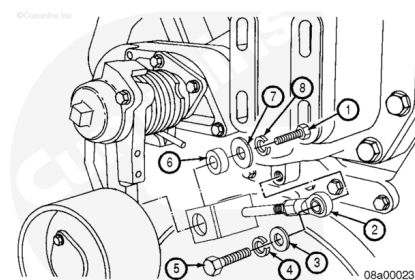
Torque Value: 60 n•m [44 ft-lb]

Install the spacer bushing (6), flat washer (7), lock washer (8), and capscrew (1) in the lower end of the control rod (2).

Install the lower end of the control rod on the fan idler arm.

Tighten the capscrew.

Torque Value: 60 n•m [44 ft-lb]



Industrial System

Install the industrial tensioner.

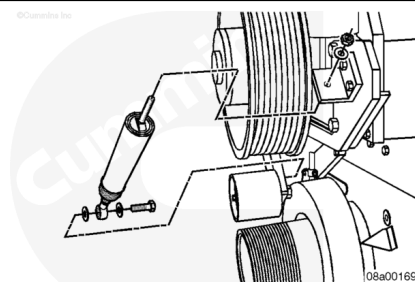
Slide the threaded-rod end of the tensioner through the hole in the upper anchor bracket and secure with a washer and nut.

Do **not** tighten the nut. It will be tightened later to adjust the belt tension.

Attach the lower clevis end of the tensioner to the idler arm with capscrews and washers.

Tighten the capscrews.

Torque Value: 92 n•m [68 ft-lb]



Adjust

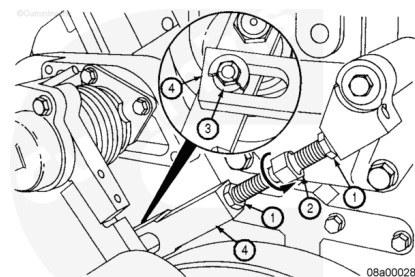
Power Generation

Note : One of the nuts has left-handed threads.

Note : There is no adjustment required for engines equipped with a shock absorber.

The fan belt **must** be installed and under the tension of the fan idler arm spring to adjust the control rod.

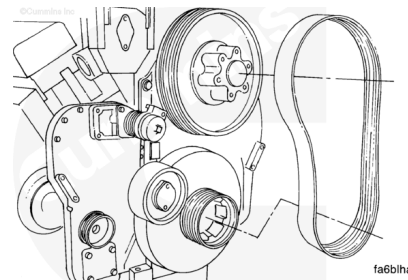
The fan belt and a portion of the flat washer are **not** shown for clarity.



Turn the adjusting screw (2) until the end of the slot on the lower control rod end (4) is touching the spacer (3).

Hold the adjusting screw and tighten the two jam nuts (1).

Install the fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/57/57-008-002.html>).



Industrial System

The fan belt is tensioned at the threaded rod end of the tensioner at the upper anchor bracket.

To adjust tension, loosen the jam nut (2) and adjust to specification with the tensioning nut (1).

Tighten the jam nut.

Use service tool belt tension gauge for industrial engines, Part Number 3164750, to adjust the belt tension.

The fan belt tension specification is 2669 N [600 lbf].

